

Module 2

SECTION 7.4: INTEGRATION OF RATIONAL FUNCTIONS BY PARTIAL FRACTIONS

As mentioned in the first paragraph in the text, the method of partial fractions is based on *reversing* the addition of two or more fractions. Consider

$$\begin{aligned}\frac{1}{x+2} + \frac{3}{x-4} &= \frac{x-4 + 3(x+2)}{(x+2)(x-4)} \\ &= \frac{4x+2}{(x+2)(x-4)}\end{aligned}$$

This is just the normal addition of two fractions. Next consider the integral

$$\int \frac{4x+2}{(x+2)(x-4)} dx$$

None of the methods we have covered apply to this form. Instead we can write

$$\int \frac{4x+2}{(x+2)(x-4)} dx = \int \left(\frac{1}{x+2} + \frac{3}{x-4} \right) dx$$

and perform the integration on the *partial* fractions. The answer is

$$\ln|x+2| + 3\ln|x-4| + C$$

The major concern in this section is how to express a single fraction (or a rational function) as the sum of two or more simpler fractions.

FIRST CONSIDERATION

If the degree of the numerator is greater than or equal to the degree of the denominator, carry out a long division before integrating. Because you may not have used long division recently, we also show the process.

Example. Evaluate $\int \frac{x^3 - 3x}{x+2} dx$

First, use long division to divide $x+2$ into $x^3 - 3x$.

$$\begin{array}{r} x^2 \\ x+2 \overline{) x^3 - 3x} \\ \underline{x^3 + 2x^2} \\ -2x^2 \end{array}$$

A.

$$\begin{array}{r} x^2 - 2x \\ x+2 \overline{) x^3 - 3x} \\ \underline{x^3 + 2x^2} \\ -2x^2 - 3x \\ \underline{-2x^2 - 4x} \\ +x \end{array}$$

B.

$$\begin{array}{r} x^2 - 2x + 1 \\ x+2 \overline{) x^3 - 3x} \\ \underline{x^3 + 2x^2} \\ -2x^2 - 3x \\ \underline{-2x^2 - 4x} \\ +x \\ \underline{x + 2} \\ -2 \end{array}$$

C.

The quotient is $x^2 - 2x + 1$ and the remainder is -2 . Then

$$\frac{x^3 - 3x}{x + 2} = x^2 - 2x + 1 + \frac{-2}{x + 2}$$

and

$$\begin{aligned} \int \frac{x^3 - 3x}{x + 2} dx &= \int (x^2 - 2x + 1) dx + \int \frac{-2}{x + 2} dx \\ &= \frac{x^3}{3} - \frac{2x^2}{2} + x - 2 \ln|x + 2| + C \end{aligned}$$

OVERVIEW

We give an overview of the method of partial fractions through examples with comments.

$$1. \quad \frac{5x - 4}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1}$$

We are assuming that we can find A and B such that when we add the two fractions the sum of the numerators will be $5x - 4$. We have two choices.

- (a) Multiply both sides by $(x - 2)(x + 1)$. This is the approach that is used in the text.

(b) Add the two fractions on the right and equate the numerators.

The two methods will give the same result.

$$\begin{aligned}\frac{5x - 4}{(x - 2)(x + 1)} &= \frac{A}{x - 2} \cdot \frac{x + 1}{x + 1} + \frac{B}{x + 1} \cdot \frac{x - 2}{x - 2} \\ &= \frac{A(x + 1) + B(x - 2)}{(x - 2)(x + 1)}\end{aligned}$$

For the two fractions to be equal, the numerators must be equal.

$$\begin{aligned}5x - 4 &= A(x + 1) + B(x - 2) \\ &= Ax + A + Bx - 2B \\ &= (A + B)x + (A - 2B)\end{aligned}$$

The x-coefficients must be equal,

$$A + B = 5$$

and the constants must be equal.

$$A - 2B = -4$$

Subtract the second equation from the first.

$$3B = 9$$

Then $B = 3$ and $A = 2$.

Finally,

$$\begin{aligned}\int \frac{5x - 4}{(x - 2)(x + 1)} dx &= \int \frac{2}{x - 2} dx + \int \frac{3}{x + 1} dx \\ &= 2\ln|x - 2| + 3\ln|x + 1| + C\end{aligned}$$

SECOND METHOD

The note on **page 509** describes a second method for determining the coefficients A and B in

$$5x - 4 = A(x + 1) + B(x - 2)$$

First let $x = -1$. Then

$$\begin{aligned}5(-1) - 4 &= A(0) + B(-1 - 2) \\ -9 &= -3B \\ B &= 3\end{aligned}$$

Also let $x = 2$.

$$\begin{aligned} 5(2) - 4 &= A(2 + 1) + 3(2 - 2) \\ 6 &= 3A \\ A &= 2 \end{aligned}$$

Suggestion: Try both methods and see which you prefer. In some cases a combination of the two approaches is appropriate.

$$2. \quad \frac{3x^2 + 13}{(x + 2)(x - 3)^2} = \frac{A}{x + 2} + \frac{B}{x - 3} + \frac{C}{(x - 3)^2}$$

Because the factor $x - 3$ occurs twice, we must include one fraction with the denominator $x - 3$ and *another* with the denominator $(x - 3)^2$.

We add the three fractions on the right.

$$\frac{3x^2 + 13}{(x + 2)(x - 3)^2} = \frac{A}{x + 2} \cdot \frac{(x - 3)^2}{(x - 3)^2} + \frac{B}{x - 3} \cdot \frac{(x - 3)(x + 2)}{(x - 3)(x + 2)} + \frac{C}{(x - 3)^2} \cdot \frac{x + 2}{x + 2}$$

Equating the numerators, we have

$$3x^2 + 13 = A(x - 3)^2 + B(x - 3)(x + 2) + C(x + 2)$$

Let $x = 3$.

$$\begin{aligned} 3 \cdot 3^2 + 13 &= A(3 - 3)^2 + B(3 - 3)(3 + 2) + C(3 + 2) \\ 40 &= 5C \\ C &= 8 \end{aligned}$$

Let $x = -2$.

$$\begin{aligned} 3 \cdot 4 + 13 &= A(-2 - 3)^2 + B(-2 - 3)(-2 + 2) + C(-2 + 2) \\ 25 &= 25A \\ A &= 1 \end{aligned}$$

We now arbitrarily let $x = 0$ and include $A = 1$, $C = 8$.

$$\begin{aligned} 3 \cdot 0 + 13 &= 1(0 - 3)^2 + B(0 - 3)(0 + 2) + 8(0 + 2) \\ 13 &= 9 - 6B + 16 \\ B &= 2 \end{aligned}$$

Then

$$\begin{aligned}\int \frac{3x^2 + 13}{(x+2)(x-3)^2} dx &= \int \frac{1}{x+2} dx + \int \frac{2}{x-3} dx + \int \frac{8}{(x-3)^2} dx \\ &= \ln|x+2| + 2\ln|x-3| + 8 \frac{(x-3)^{-1}}{-1} + C\end{aligned}$$

3. Both $(x-3)^2$ and $x^2 + 9$ are quadratic expressions, but they are treated differently in the method of partial fractions. Consider the integral

$$\int \frac{3x^2 - 10x + 22}{(2x+1)(x^2+9)} dx$$

The term $\frac{Bx + C}{x^2 + 9}$ must be included in the equation

$$\frac{3x^2 - 10x + 22}{(2x+1)(x^2+9)} = \frac{A}{2x+1} + \frac{Bx + C}{x^2 + 9}$$

Note the linear expression *involving* x in the *numerator* of the second fraction. In the last example we used

$$\frac{B}{x-3} + \frac{C}{(x-3)^2}$$

where both B and C were constants. In the second fraction the denominator $(x-3)^2$ is treated as a *repeated linear* factor. In contrast the quadratic, $x^2 + 9$, is not factorable and hence treated differently.

The coefficients A , B , and C are determined as before. Add the fractions on the right side.

$$\frac{3x^2 - 10x + 22}{(2x+1)(x^2+9)} = \frac{A}{2x+1} \cdot \frac{x^2+9}{x^2+9} + \frac{Bx+C}{x^2+9} \cdot \frac{2x+1}{2x+1}$$

Then

$$3x^2 - 10x + 22 = A(x^2 + 9) + (Bx + C)(2x + 1)$$

Let $x = -\frac{1}{2}$

$$\frac{3}{4} + 5 + 22 = A\left(\frac{1}{4} + 9\right)$$

Multiply by 4.

$$3 + 20 + 88 = A(1 + 36)$$

$$37A = 111$$

$$A = 3$$

Because $x^2 + 9$ doesn't equal zero for any real value of x , we switch to the method used in the text.

$$\begin{aligned} 3x^2 - 10x + 22 &= 3(x^2 + 9) + (Bx + C)(2x + 1) \\ &= 3x^2 + 27 + 2Bx^2 + Bx + 2Cx + C \\ &= (3 + 2B)x^2 + (B + 2C)x + C + 27 \end{aligned}$$

$$\begin{array}{lcl} \text{Then} & 3 + 2B = 3 & , \quad B + 2C = -10, \quad C + 27 = 22 \\ \text{and} & B = 0 & \quad \quad \quad C = -5 \end{array}$$

So

$$\frac{3x^2 - 10x + 22}{(2x + 1)(x^2 + 9)} = \frac{3}{2x + 1} + \frac{-5}{x^2 + 9}$$

and

$$\begin{aligned} \int \frac{3x^2 - 10x + 22}{(2x + 1)(x^2 + 9)} dx &= \int \frac{3}{2x + 1} dx + \int \frac{-5}{x^2 + 9} dx \\ &= \frac{3}{2} \ln|2x + 1| - \frac{5}{3} \tan^{-1}\left(\frac{x}{3}\right) + C \end{aligned}$$

To evaluate the second integral use **(10)** on *page 511*.

$$(10) \quad \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

NOTE

Instead of switching methods midstream, we could have selected two arbitrary values of x . Two convenient values are 0 and 1.

Let $x = 0$ in

$$\begin{aligned} 3x^2 - 10x + 22 &= 3(x^2 + 9) + (Bx + C)(2x + 1) \\ 22 &= 27 + C \\ C &= -5 \end{aligned}$$

Let $x = 1$.

$$\begin{aligned}
 3 - 10 + 22 &= 30 + (B - 5)(3) \\
 15 &= 30 + 3B - 15 \\
 B &= 0
 \end{aligned}$$

4. The last case involves a repeated quadratic factor as shown in **(11)** on **page 513**. The numerator continues to be of the form $Ax + B$. This follows the pattern in part (2) above as shown in

$$\frac{A}{x + 1} + \frac{B}{(x + 1)^2}$$

and

$$\frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}$$

5. The note on **page 513** indicates how to evaluate

$$\int \frac{Ax + B}{x^2 + 1} dx$$

First rewrite it as two separate integrals.

$$\int \frac{Ax + B}{x^2 + 1} dx = \int \frac{Ax}{x^2 + 1} dx + \int \frac{B}{x^2 + 1} dx$$

Then note that

$$\int \frac{Ax}{x^2 + 1} dx = \frac{A}{2} \int \frac{2x}{x^2 + 1} dx = \frac{A}{2} \ln|x^2 + 1| + C$$

and

$$\int \frac{B}{x^2 + 1} dx = \frac{B}{1} \tan^{-1}\left(\frac{x}{1}\right) + C$$

COMMENT ON THEORETICAL JUSTIFICATIONS

Part 1 above involved two sets of equations

$$(1) \quad \frac{5x - 4}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1}$$

$$(2) \quad 5x - 4 = A(x + 1) + B(x - 2)$$

We let $x = -1$, as the first step in solving for A, B and C. You may have noticed that equation (1) is undefined when $x = -1$. This difficulty is discussed in the text in the note on **page 509** and also in **exercise 75, page 516**. Although equation (2) was derived using equation (1), it has an existence separate from the original. Think of equation (2) as equating two polynomials. The polynomial $5x - 4$ is equal to another polynomial, $A(x + 1) + B(x - 2)$, where A and B are constants.

Then remember that *all* polynomials are continuous for all values of x . Equation (1) is not continuous at $x = -1$ and $x = 2$ (its graph has vertical asymptotes at $x = -1$ and $x = 2$) but the polynomials in equation (2) are continuous (no breaks in their graphs). So they will have values for -1 and 2. Then our methods for determining A and B seem reasonable.

However, some things that are reasonable turn out not to be true. **Exercise 75**, goes further and outlines a proof that equation (2) is true for all values of x . The proof is more abstract because it applies to all types of polynomials that might appear in this procedure. The key step is to look at the following limits. (We consider only a particular case instead of the general situation.)

$$\lim_{x \rightarrow 2} (5x - 4) \text{ and } \lim_{x \rightarrow 2} (A(x + 1) + B(x - 2))$$

Consider x approaching 2 using the values 2.1, 2.01, 2,001, *etc.* .

For each of these values we know that the two polynomials are equal. So the claim is that the limits must be equal because the polynomials are continuous at $x = 2$. To agree with this last claim, slowly work through the process of taking a limit, and remember that the polynomials are equal at 2.1, 2.01, 2,001, *etc.*

RATIONALIZING SUBSTITUTIONS

The last method of integration involves a special substitution.

In **Example 9, page 515**, the substitution is $u = \sqrt{x + 4}$.

In this case we have the square root of a linear expression. Contrast this with $\sqrt{x^2 + 4}$, the square root of a quadratic expression. Hopefully you recall that a trig substitution eliminates the square root. The type of substitution used in this section does

not work for $\sqrt{x^2 + 4}$ because $u = \sqrt{x^2 + 4}$ leads to

$$\begin{aligned} u^2 &= x^2 + 4 \\ u^2 - 4 &= x^2 \\ x &= \sqrt{u^2 - 4} \end{aligned}$$

The square root would *not* be eliminated if a term containing x is present.

In contrast, for $u = \sqrt{x + 4}$

$$u^2 = x + 4$$

and

$$x = u^2 - 4$$

$$dx = 2u \, du$$

So in **Example 9**, the square root is eliminated without other complications. Note that there are three components to the substitution.

Replace $\sqrt{x + 4}$ with u
 x with $u^2 - 4$
 and dx with $2u \, du$.

In **Exercise 47**, the substitution, $u = \sqrt[6]{x}$, will eliminate both the square root and the cube root.

Note:

$$\begin{aligned} u &= \sqrt[6]{x} = x^{1/6} \\ \sqrt{x} &= x^{1/2} = x^{3/6} = u^3 \\ \sqrt[3]{x} &= x^{1/3} = x^{2/6} = u^2 \end{aligned}$$

The exponents can be written in terms of the common denominator 6.

SECTION 7.5: STRATEGY FOR INTEGRATION

In some sense this section is a test of how well you have absorbed the methods of integration presented in this chapter. Note the admonition that the first 18 entries in the table on **page 517** should be memorized. You now know that some of the methods of integration are complex, and it is of great help to recognize basic integrals when they occur. Having this information in your head makes a big difference from having to refer to a list. A goal in this chapter is to develop your *perception* of the forms that indicate which method to use. If information is in your head, it is more likely

step away from your perceptive process and requires greater mental effort on your part. To gain the feeling that things are really flowing, we recommend that you memorize the 18 entries in the table.

Next carefully read the 4 steps presented on **pages 518-19**. In particular, look at step 3. It contains a good summary and also suggests forms to look for. Forms matching integration by parts present the greatest challenge. Again we suggest that you look at the types of problems on **pages 490 and 491 (Sec. 7.1)** and that you also briefly review the four patterns covered in Module 1. Expect, however, that a new integral may appear, and the only way to recognize that integration by parts works is to try it.

It is important to read ***Can We Integrate All Continuous Functions?*** on **pp. 520-21**: realize that the six integrals listed *there* cannot be integrated using *elementary functions*. The methods we have covered do not work for all integrals. Later in this chapter we cover tables of integrals and numerical approximations. You will not find those six integrals in a table of integrals. However, the procedures involving numerical approximations apply to *all* integrals and, hence, to these six integrals. Also note the reference to *infinite series*, a topic that is covered later in this course.